

Network Forensic

❖ Introduction to Network

- ⇒ A network is a system that connects multiple devices together, allowing them to exchange information and share resources using communication protocols.

🌐 Network Types:

1. Personal Area Network (PAN)

A Personal Area Network (PAN) connects personal devices within a short distance (up to 10 meters).

Example: Smartphone connected to Bluetooth earbuds or a smartwatch.

2. Local Area Network (LAN)

A Local Area Network (LAN) connects devices within a small area such as a home, school, office, or computer lab. It provides high-speed communication and resource sharing.

Example: A college computer laboratory.

3. Campus Area Network (CAN)

A Campus Area Network (CAN) connects multiple LANs within a campus or organization.

Example: A university connecting its departments, library, and administration building.

4. Metropolitan Area Network (MAN)

A Metropolitan Area Network (MAN) connects networks across a city.

Example: A bank connecting all its branches within the same city.

5. Wide Area Network (WAN)

A Wide Area Network (WAN) connects networks over large geographical areas, such as countries or continents.

Example: The Internet.

✚ Network Topologies:

- Bus, Star, Ring, Mesh, Tree, Hybrid

✚ Network Devices:

- Router, Switch, Hub, Bridge, Repeater, Gateway, Firewall, Wireless Access Point, Modem, Network Interface Card (NIC)

✚ OSI Model Process:

1	Application Layer Allows users to access network services like web browsing and email.
2	Presentation Layer Translates, encrypts, and compresses data for application use.
3	Session Layer Creates, manages, synchronizes, and ends communication sessions.
4	Transport Layer Ensures reliable delivery by dividing data into segments.
5	Network Layer Determines the best path for data using IP addresses.
6	Data Link Layer Transfers data between devices on the same local network.
7	Physical Layer Converts data into electrical, optical, or radio signals.

Tcp Model Process :

1	Network Access Layer Sends data through Wi-Fi, Ethernet, or cable, like a road carrying a parcel.
2	Internet Layer Finds the destination using the IP address, like a home address on a parcel.
3	Transport Layer Breaks data into small parts and delivers it correctly, like dividing a book into pages.
4	Application Layer Lets users use internet applications like Chrome, WhatsApp, Gmail, and YouTube.

❖ **TCPDUMP**

Defination:

- TCPDUMP is a network monitoring tool that provides visibility into the traffic activity taking place on a network.
- Software that displays the output in human readable format
- tcpdump → Works on Linux/Unix.
- WinDump → Works on Windows.

Installation

Method 1: In the Kali Linux, Ubuntu, or Debian terminal, use the following commands.

```
sudo apt update
```

```
sudo apt install tcpdump
```

```
tcpdump -version
```

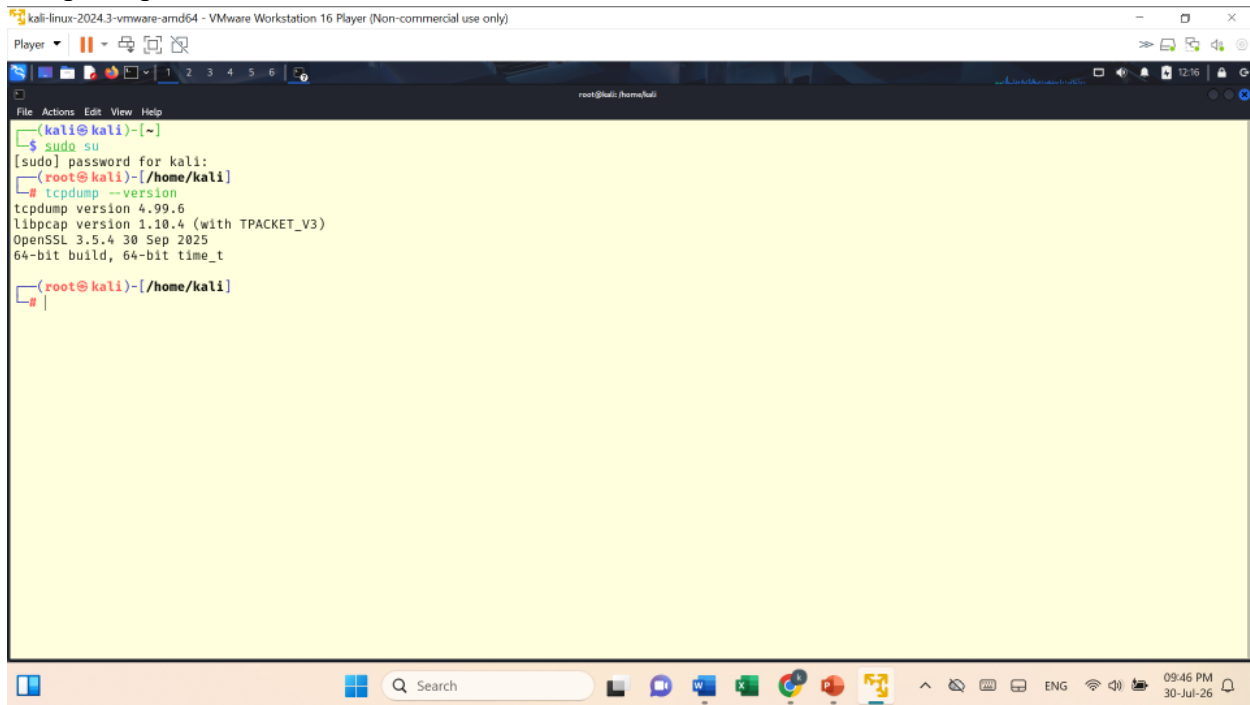
Method 2: Red Hat / CentOS / Fedora

```
sudo apt update
```

```
sudo dnf install tcpdump
```

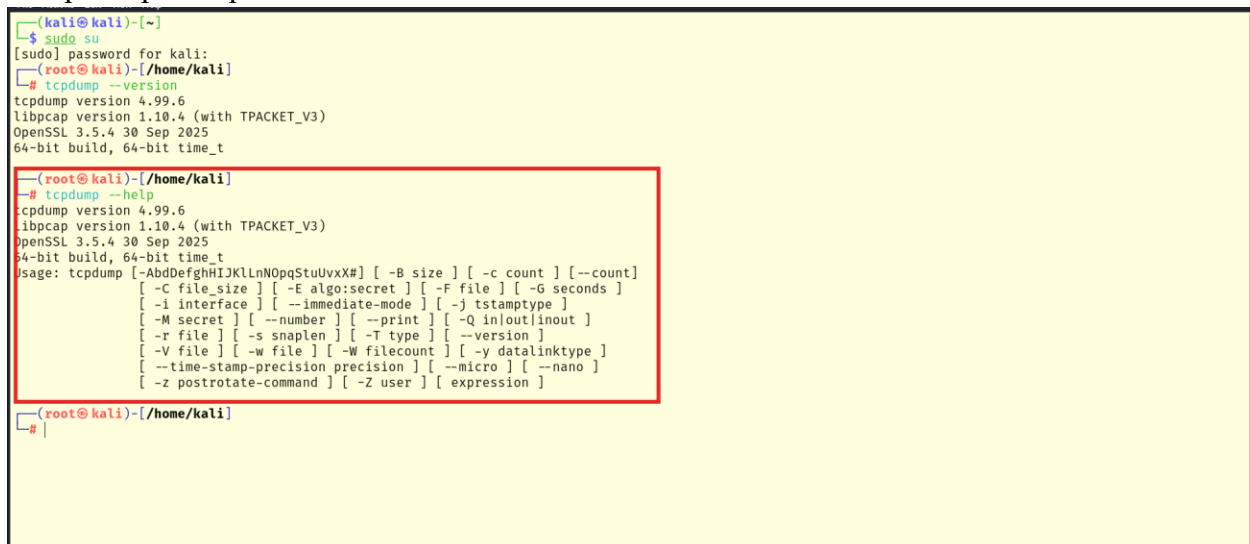
Practical: (TCPDUMP) I perform Using Kali Linux.

1. tcpdump --version



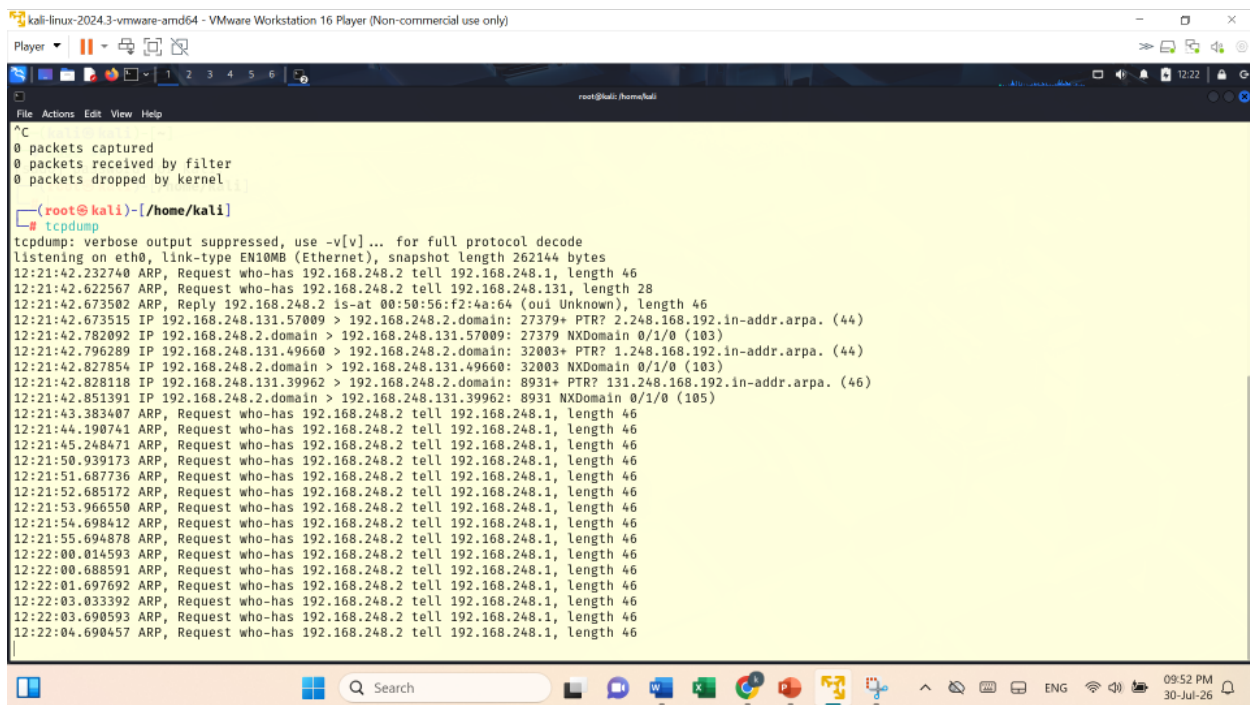
```
kali-linux-2024.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
root@kali: /home/kali
File Actions Edit View Help
(kali@kali)~$ sudo su
[sudo] password for kali:
(root@kali)~$ tcpdump --version
tcpdump version 4.99.6
libpcap version 1.10.4 (with TPACKET_V3)
OpenSSL 3.5.4 30 Sep 2025
64-bit build, 64-bit time_t
(root@kali)~$
```

2. tcpdump --help



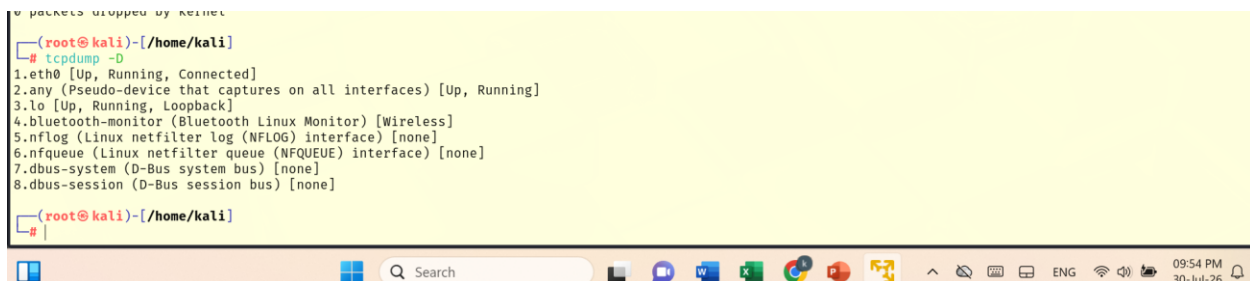
```
(kali@kali)~$ sudo su
[sudo] password for kali:
(root@kali)~$ tcpdump --version
tcpdump version 4.99.6
libpcap version 1.10.4 (with TPACKET_V3)
OpenSSL 3.5.4 30 Sep 2025
64-bit build, 64-bit time_t
(root@kali)~$ tcpdump --help
tcpdump version 4.99.6
libpcap version 1.10.4 (with TPACKET_V3)
OpenSSL 3.5.4 30 Sep 2025
64-bit build, 64-bit time_t
Usage: tcpdump [-AbdDefghHIJKLlnNOpqStuUvXx#] [-B size] [-c count] [--count]
      [-C file_size] [-E algo:secret] [-F file] [-G seconds]
      [-i interface] [--immediate-mode] [-j tstamptype]
      [-M secret] [--number] [--print] [-Q in|out|inout]
      [-r file] [-s snaplen] [-T type] [--version]
      [-V file] [-w file] [-W filecount] [-y datalinktype]
      --time-stamp-precision precision [-micro] [--nano]
      [-z postrotate-command] [-Z user] [expression]
```

3. tcpdump: Capture traffic on lowest numbered network interface



```
kali-linux-2024.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
File Actions Edit View Help
root@kali: /home/kali
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
root@kali) ~
# tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:21:42.232740 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:42.622567 ARP, Request who-has 192.168.248.2 tell 192.168.248.131, length 28
12:21:42.673502 ARP, Reply 192.168.248.2 is-at 00:50:56:f2:4a:64 (oui Unknown), length 46
12:21:42.673515 IP 192.168.248.131.57009 > 192.168.248.2.domain: 27379+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:21:42.782092 IP 192.168.248.2.domain > 192.168.248.131.57009: 27379 NXDomain 0/1/0 (103)
12:21:42.796289 IP 192.168.248.131.49660 > 192.168.248.2.domain: 32003+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:21:42.827854 IP 192.168.248.2.domain > 192.168.248.131.49660: 32003 NXDomain 0/1/0 (103)
12:21:42.828118 IP 192.168.248.131.39962 > 192.168.248.2.domain: 8931+ PTR? 131.248.168.192.in-addr.arpa. (46)
12:21:42.851391 IP 192.168.248.2.domain > 192.168.248.131.39962: 8931 NXDomain 0/1/0 (105)
12:21:43.383407 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:44.190741 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:45.248471 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:50.939173 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:51.687736 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:52.685172 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:53.966550 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:54.698412 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:21:55.694878 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:22:00.814593 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:22:00.688591 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:22:01.697692 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:22:03.833392 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:22:03.690593 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:22:04.690457 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

4. tcpdump -D: It is used to display (list) all available network interfaces on your computer.



```
0 packets dropped by kernel
root@kali) ~
# tcpdump -D
1.eth0 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
root@kali) ~
```

5. tcpdump -i 2: Based on '-D' interface output, you can use interface number. In this example we will capture traffic on 'loopback' interface (order number 2)

```
(root@kali)-[/home/kali]
# tcpdump -d
1.eth0 [Up, Running, Connected]
2.any [Pseudo-device that captures on all interfaces] [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor [Bluetooth Linux Monitor] [Wireless]
5.nflog [Linux netfilter log (NFLOG interface)] [none]
6.nfqueue [Linux netfilter queue (NFQUEUE interface)] [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]

(root@kali)-[/home/kali]
# tcpdump -i 2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
12:27:16.199065 eth0 B IP 192.168.248.1.netbios-dgm > 192.168.248.255.netbios-dgm: UDP, length 201
12:27:16.239476 eth0 Out IP 192.168.248.131.47694 > 192.168.248.2.domain: 16663+ PTR? 255.248.168.192.in-addr.arpa. (46)
12:27:16.305903 eth0 B ARP, Request who-has 192.168.248.131 tell 192.168.248.2, length 46
12:27:16.305920 eth0 Out ARP, Reply 192.168.248.131 is-at 00:0c:29:ec:3b:3f (ok Unknown), length 28
12:27:16.329810 eth0 In IP 192.168.248.2.domain > 192.168.248.131.47694: 16663 NXDomain 0/1/0 (105)
12:27:16.321303 eth0 Out IP 192.168.248.131.41543 > 192.168.248.2.domain: 15857+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:27:16.396092 eth0 In IP 192.168.248.2.domain > 192.168.248.131.41543: 15857 NXDomain 0/1/0 (103)
12:27:16.396455 eth0 Out IP 192.168.248.131.46252 > 192.168.248.2.domain: 16044+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:27:16.443272 eth0 In IP 192.168.248.2.domain > 192.168.248.131.46252: 16044 NXDomain 0/1/0 (103)
12:27:16.443523 eth0 Out IP 192.168.248.131.45329 > 192.168.248.2.domain: 44165+ PTR? 131.248.168.192.in-addr.arpa. (46)
12:27:16.499739 eth0 In IP 192.168.248.2.domain > 192.168.248.131.45329: 44165 NXDomain 0/1/0 (105)
12:27:16.506823 eth0 In ARP, Request who-has 192.168.248.1 tell 192.168.248.131, length 28
12:27:21.321441 eth0 In ARP, Reply 192.168.248.1 is-at 00:50:56:f2:4a:64 (ok Unknown), length 46
12:27:29.201454 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:27:30.188193 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

6. `tcpdump -i eth0`: captures and displays all network packets passing through the eth0 network interface.

[illegible]

7. `tcpdump -i any`: captures packets from all available network interfaces at the same time.

```
(root@kali)-[/home/kali]
# tcpdump -i any
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
12:31:46.878907 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:31:46.951799 eth0 Out IP 192.168.248.131.34748 > 192.168.248.2.domain: 8798+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:31:46.980004 eth0 In IP 192.168.248.2.domain > 192.168.248.131.34748: 8798 NXDomain 0/0/0 (44)
12:31:46.981229 eth0 Out IP 192.168.248.131.34053 > 192.168.248.2.domain: 30665+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:31:46.986770 eth0 In IP 192.168.248.2.domain > 192.168.248.131.34053: 30665 NXDomain 0/0/0 (44)
12:31:47.055474 eth0 Out IP 192.168.248.131.35403 > 192.168.248.2.domain: 44263+ PTR? 131.248.168.192.in-addr.arpa. (46)
12:31:47.061516 eth0 In IP 192.168.248.2.domain > 192.168.248.131.35403: 44263 NXDomain 0/0/0 (46)
12:31:47.702398 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:31:48.699407 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:31:49.900580 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:31:50.697440 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:31:51.695150 eth0 B ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

```
(root@kali)-[/home/kali]
# tcpdump -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:37:29.194139 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:30.286479 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:31.192595 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:32.203621 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:37.841499 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:38.702593 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:39.721797 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:40.856975 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:41.699115 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:42.696681 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:46.901052 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:47.696419 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:48.690381 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:49.925808 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:50.703738 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:37:51.700742 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:39:13.260368 IP 192.168.248.1.138 > 192.168.248.255.138: UDP, length 201
```

```
kali-linux-2024-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player | [Icons] | [Progress Bar] | 1 2 3 4 5 6 | [Volume Icon]
root@kali: /home/kali
File Actions Edit View Help
^C
122 packets captured
122 packets received by filter
0 packets dropped by kernel

(root@kali)-[/home/kali]
# tcpdump -v
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:45:26.017489 IP (tos 0x0, ttl 4, id 2277, offset 0, flags [none], proto UDP (17), length 470)
    192.168.248.1.1900 > 239.255.255.250:1900: UDP, length 442
12:45:26.021137 IP6 (flowlabel 0xe0430, hlim 4, next-header UDP (17), payload length 458) fe80::a3a8:cb23:5977:3ffe.1900 > ff02::c.1900: [udp sum ok] UDP, le
ngh: 458
12:45:26.059314 IP (tos 0x0, ttl 64, id 24767, offset 0, flags [DF], proto UDP (17), length 74)
    192.168.248.131.60826 > 192.168.248.2.domain: 61778+ PTR? 250.255.255.250.in-addr.arpa. (66)
12:45:26.089271 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.131 tell 192.168.248.2, length 46
12:45:26.089289 ARP, Ethernet (len 6), IPv4 (len 4), Reply 192.168.248.131 is-at 00:0c:29:ec:3b:3f (oui Unknown), length 28
12:45:26.089734 IP (tos 0x0, ttl 128, id 65262, offset 0, flags [none], proto UDP (17), length 131)
    192.168.248.2.domain > 192.168.248.131.60826: 61778 NXDomain 0/1/0 (103)
12:45:26.090283 IP (tos 0x0, ttl 64, id 11756, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.248.131.51343 > 192.168.248.2.domain: 53161+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:45:26.111132 IP (tos 0x0, ttl 128, id 65263, offset 0, flags [none], proto UDP (17), length 131)
    192.168.248.2.domain > 192.168.248.131.51343: 53161 NXDomain 0/1/0 (103)
12:45:26.111495 IP (tos 0x0, ttl 64, id 18614, offset 0, flags [DF], proto UDP (17), length 118)
    192.168.248.131.48796 > 192.168.248.2.domain: 47624+ PTR? c.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.f.f.ip6.arpa. (90)
12:45:26.142954 IP (tos 0x0, ttl 128, id 65264, offset 0, flags [none], proto UDP (17), length 182)
    192.168.248.2.domain > 192.168.248.131.48796: 47624 NXDomain 0/1/0 (154)
12:45:26.143189 IP (tos 0x0, ttl 64, id 22059, offset 0, flags [DF], proto UDP (17), length 118)
    192.168.248.131.50827 > 192.168.248.2.domain: 28788+ PTR? e.f.f.3.7.7.9.5.3.2.b.c.c.a.0.0.0.0.0.0.0.0.0.0.0.e.f.ip6.arpa. (90)
12:45:26.173712 IP (tos 0x0, ttl 128, id 65265, offset 0, flags [none], proto UDP (17), length 177)
    192.168.248.2.domain > 192.168.248.131.50827: 28788 NXDomain 0/1/0 (149)
12:45:26.174739 IP (tos 0x0, ttl 64, id 34430, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.248.131.33066 > 192.168.248.2.domain: 25018+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:45:26.205517 IP (tos 0x0, ttl 128, id 65266, offset 0, flags [none], proto UDP (17), length 131)
    192.168.248.2.domain > 192.168.248.131.33066: 25018 NXDomain 0/1/0 (103)
```

```

root@kali:~/home/kali# tcpdump -vv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:51:31.698007 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.1 tell 192.168.248.1, length 46
12:51:31.727541 IP (tos 0x0, ttl 64, id 23376, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.248.131.55099 > 192.168.248.2.domain: [bad udp cksum 0x721d -> 0x17c31] 56260+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:51:31.793534 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.131 tell 192.168.248.2, length 46
12:51:31.793549 ARP, Ethernet (len 6), IPv4 (len 4), Reply 192.168.248.131 is-at: 00:0c:29:ec:3b:3f (oui Unknown), length 28
12:51:31.793727 IP (tos 0x0, ttl 128, id 65275, offset 0, flags [none], proto UDP (17), length 131)
    192.168.248.2.domain > 192.168.248.131.55099: [udp sum ok] 56260 NXDomain q: PTR? 2.248.168.192.in-addr.arpa. 0/1/0 ns: 168.192.in-addr.arpa. SOA localho
st. nobody.invalid. 1 3600 1200 604800 10800 (103)
12:51:31.793982 IP (tos 0x0, ttl 64, id 16281, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.248.131.42996 > 192.168.248.2.domain: [bad udp cksum 0x721d -> 0x689f1] 47664+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:51:31.858116 IP (tos 0x0, ttl 128, id 65276, offset 0, flags [none], proto UDP (17), length 131)
    192.168.248.2.domain > 192.168.248.131.42996: [udp sum ok] 47664 NXDomain q: PTR? 1.248.168.192.in-addr.arpa. 0/1/0 ns: 168.192.in-addr.arpa. SOA localho
st. nobody.invalid. 1 3600 1200 604800 10800 (103)
12:51:31.858646 IP (tos 0x0, ttl 64, id 772, offset 0, flags [DF], proto UDP (17), length 74)
    192.168.248.131.50561 > 192.168.248.2.domain: [bad udp cksum 0x721f -> 0x6a801] 25997+ PTR? 131.248.168.192.in-addr.arpa. (46)

```

11. `tcpdump -vvv`: `tcpdump -vvv` is used to display the maximum available details about captured network packets for in-depth network analysis and troubleshooting.

```
root@kali:~# sudo su
[sudo] password for kali:
root@kali:~# tcpdump -vvv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^12:51:36.922659 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.131, length 28
12:51:36.931130 IP (tos 0x0, ttl 64, id 59108, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.248.131.59561 > 192.168.248.2.domain: [bad udp cksum 0x721d -> 0xa185!] 16532+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:51:36.954424 ARP, Ethernet (len 6), IPv4 (len 4), Reply 192.168.248.2 is-at 00:50:56:f2:4a:64 (oui Unknown), length 46
12:51:36.961593 IP (tos 0x0, ttl 128, id 65278, offset 0, flags [none], proto UDP (17), length 72)
    192.168.248.2.domain > 192.168.248.131.59561: [udp sum ok] 16532 NXDomain q: PTR? 2.248.168.192.in-addr.arpa. 0/0/0 (44)
12:51:36.961871 IP (tos 0x0, ttl 64, id 9461, offset 0, flags [DF], proto UDP (17), length 74)
    192.168.248.131.60814 > 192.168.248.2.domain: [bad udp cksum 0x721f -> 0x6c26!] 15322+ PTR? 131.248.168.192.in-addr.arpa. (46)
12:51:36.968971 IP (tos 0x0, ttl 128, id 65279, offset 0, flags [none], proto UDP (17), length 74)
    192.168.248.2.domain > 192.168.248.131.60814: [udp sum ok] 15322 NXDomain q: PTR? 131.248.168.192.in-addr.arpa. 0/0/0 (46)
12:51:40.295237 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:40.370941 IP (tos 0x0, ttl 64, id 26814, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.248.131.47818 > 192.168.248.2.domain: [bad udp cksum 0x721d -> 0x6d72!] 41607+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:51:40.378230 IP (tos 0x0, ttl 128, id 65280, offset 0, flags [none], proto UDP (17), length 72)
    192.168.248.2.domain > 192.168.248.131.47818: [udp sum ok] 41607 NXDomain q: PTR? 1.248.168.192.in-addr.arpa. 0/0/0 (44)
12:51:41.200898 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:42.194713 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:43.318342 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:44.189791 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:45.205481 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:49.359371 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:50.200966 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:51.198691 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:52.384745 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:51:53.103072 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

12. `tcpdump host 192.168.248.131`: captures only the packets where the source or destination IP address is 192.168.248.131.

```
root@kali:~# tcpdump host 192.168.248.131
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

13. `tcpdump dst host 8.8.8.8`: captures only packets whose destination is 8.8.8.8.

```
root@kali:~# tcpdump dst host 8.8.8.8
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

14. `tcpdump src host 192.168.248.131`: Capture only packets where the SOURCE IP address is 192.168.248.131.

```
(root@kali)-[/home/kali]
# tcpdump src host 192.168.248.131
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:04:21.083830 IP 192.168.248.131.59862 > pnbomb-bk-in-f3.1e100.net.http: Flags [.], ack 356021761, win 31018, length 0
13:04:21.160896 IP 192.168.248.131.55165 > 192.168.248.2.domain: 64642+ PTR? 227.221.251.142.in-addr.arpa. (46)
13:04:21.188141 IP 192.168.248.131.35558 > 192.168.248.2.domain: 37632+ PTR? 131.248.168.192.in-addr.arpa. (46)
13:04:21.218781 IP 192.168.248.131.52994 > 192.168.248.2.domain: 12132+ PTR? 2.248.168.192.in-addr.arpa. (44)
13:04:25.434739 IP 192.168.248.131.45916 > lcbomp-in-f94.1e100.net.http: Flags [.], ack 1320399390, win 31017, length 0
13:04:25.466907 IP 192.168.248.131.37755 > 192.168.248.2.domain: 14030+ PTR? 94.211.178.192.in-addr.arpa. (45)
13:04:25.690711 IP 192.168.248.131.42418 > 172.66.2.5.http: Flags [.], ack 1332890013, win 31720, length 0
13:04:25.779171 IP 192.168.248.131.37192 > 192.168.248.2.domain: 32400+ PTR? 5.2.66.172.in-addr.arpa. (41)
```

15. tcpdump net 192.168.248.0/24: Captures all network traffic from the 192.168.248.0/24

```
# tcpdump net 192.168.248.0/24
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:46:40.705873 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:46:40.859478 ARP, Request who-has 192.168.248.2 tell 192.168.248.131, length 28
12:46:40.890437 ARP, Reply 192.168.248.2 is-at 00:50:56:f2:4a:64 (oui Unknown), length 46
12:46:40.890452 IP 192.168.248.131.53377 > 192.168.248.2.domain: 20208+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:46:41.159641 IP 192.168.248.2.domain > 192.168.248.131.53377: 20208 NXDomain 0/1/0 (103)
12:46:41.182723 IP 192.168.248.131.40620 > 192.168.248.2.domain: 4339+ PTR? 1.248.168.192.in-addr.arpa. (44)
```

16. tcpdump net dst 192.168.248.0/24: Captures packets whose destination IP belongs to the 192.168.248.0/24 network.

```
(root@kali)-[/home/kali]
# tcpdump dst net 192.168.248.0/24
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:52:40.695923 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:40.737781 IP 192.168.248.131.45862 > 192.168.248.2.domain: 27159+ PTR? 2.248.168.192.in-addr.arpa. (44)
12:52:40.748927 ARP, Request who-has 192.168.248.131 tell 192.168.248.2, length 46
12:52:40.748945 ARP, Reply 192.168.248.131 is-at 00:0c:29:ec:3b:3f (oui Unknown), length 28
12:52:40.766547 IP 192.168.248.2.domain > 192.168.248.131.45862: 27159 NXDomain 0/0/0 (44)
12:52:40.766883 IP 192.168.248.131.48693 > 192.168.248.2.domain: 40699+ PTR? 1.248.168.192.in-addr.arpa. (44)
12:52:40.771607 IP 192.168.248.2.domain > 192.168.248.131.48693: 40699 NXDomain 0/0/0 (44)
12:52:40.805939 IP 192.168.248.131.54399 > 192.168.248.2.domain: 26395+ PTR? 131.248.168.192.in-addr.arpa. (46)
12:52:40.809990 IP 192.168.248.2.domain > 192.168.248.131.54399: 26395 NXDomain 0/0/0 (46)
12:52:42.007843 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:42.722226 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:43.694815 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:45.741518 ARP, Request who-has 192.168.248.2 tell 192.168.248.131, length 28
12:52:45.741774 ARP, Reply 192.168.248.2 is-at 00:50:56:f2:4a:64 (oui Unknown), length 46
12:52:49.562226 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:50.186098 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:51.208184 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
12:52:52.576939 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

17. tcpdump ip6: To capture IP6 traffic, use 'ip6' protocol.

18. tcpdump udp: Captured Udp packets.

19. tcpdump port 53: Capture all packets on specified port. This will capture all protocols on port 53.

20. tcpdump not port 53: Captures all network traffic except packets that use port 53 (DNS traffic).

```
(root@kali)-[/home/kali]
# tcpdump not port 53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:09:45.461842 IP 192.168.248.131.bootpc > 192.168.248.254.bootps: BOOTP/DHCP, Request from 00:0c:29:ec:3b:3f (oui Unknown), length 282
13:09:45.462378 IP 192.168.248.254.bootps > 192.168.248.131.bootpc: BOOTP/DHCP, Reply, length 300
13:09:50.509543 ARP, Request who-has 192.168.248.2 tell 192.168.248.131, length 28
13:09:50.509759 ARP, Request who-has 192.168.248.254 tell 192.168.248.131, length 28
13:09:50.560241 ARP, Reply 192.168.248.2 is-at 00:50:56:f2:4a:64 (oui Unknown), length 46
13:09:50.576191 ARP, Reply 192.168.248.254 is-at 00:50:56:ec:0b:09 (oui Unknown), length 46
```

21. `tcpdump -t`: Captures network packets without displaying the timestamp (time) at the beginning of each packet.

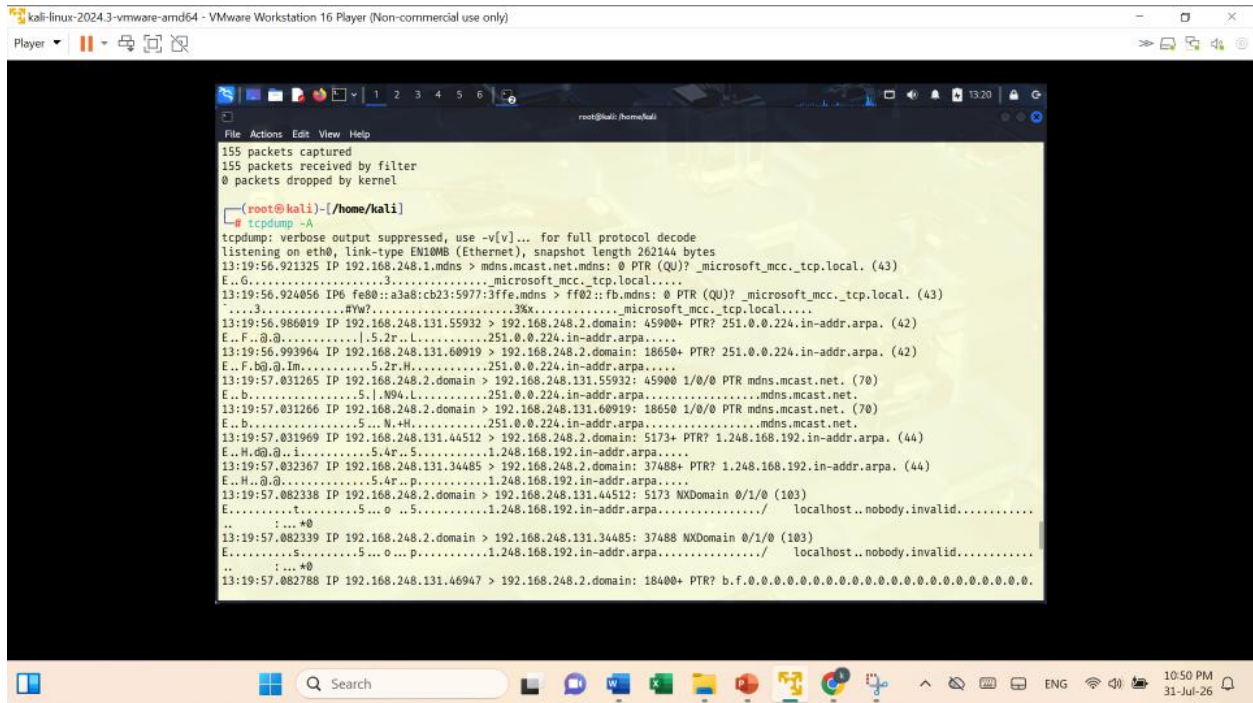
```
(root@kali)-[/home/kali]
# tcpdump -t
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
IP 192.168.248.131.58661 > 192.168.248.2.domain: 54277+ PTR? 2.248.168.192.in-addr.arpa. (44)
ARP, Request who-has 192.168.248.131 tell 192.168.248.2, length 46
ARP, Reply 192.168.248.131 is-at 00:0c:29:ec:3b:3f (oui Unknown), length 28
IP 192.168.248.2.domain > 192.168.248.131.58661: 54277 NXDomain 0/1/0 (103)
IP 192.168.248.131.33403 > 192.168.248.2.domain: 51406+ PTR? 1.248.168.192.in-addr.arpa. (44)
IP 192.168.248.2.domain > 192.168.248.131.33403: 51406 NXDomain 0/1/0 (103)
IP 192.168.248.131.42628 > 192.168.248.2.domain: 43209+ PTR? 131.248.168.192.in-addr.arpa. (46)
IP 192.168.248.2.domain > 192.168.248.131.42628: 43209 NXDomain 0/1/0 (105)
ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

22. `tcpdump -tt`: Displays each captured packet with the full Unix timestamp (seconds since 1 January 1970) for precise timing analysis.

Also available `-t` to `-ttttt`.

```
(root@kali)-[/home/kali]
# tcpdump -tt
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
1785518040.580338 IP 192.168.248.1.1900 > 239.255.255.250.1900: UDP, length 442
1785518040.581604 IP6 fe80::a3a8:cb23:5977:3ffe.1900 > ff02::c.1900: UDP, length 450
1785518040.589120 IP 192.168.248.1.1900 > 239.255.255.250.1900: UDP, length 498
1785518040.590003 IP6 fe80::a3a8:cb23:5977:3ffe.1900 > ff02::c.1900: UDP, length 506
1785518040.591248 IP 192.168.248.1.1900 > 239.255.255.250.1900: UDP, length 451
1785518040.592678 IP6 fe80::a3a8:cb23:5977:3ffe.1900 > ff02::c.1900: UDP, length 459
1785518040.676870 IP 192.168.248.131.40473 > 192.168.248.2.domain: 19487+ PTR? 250.255.255.239.in-addr.arpa. (46)
1785518040.857099 IP 192.168.248.131.33168 > 192.168.248.2.domain: 18982+ PTR? 250.255.255.239.in-addr.arpa. (46)
```

23. `tcpdump -A`: Print captured packets in ASCII format.



24. `tcpdump -x`: Print captured packets in Hex format.

```

(root@kali)-[/home/kali]
# tcpdump -x
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:19:56.921322 IP 192.168.248.1.mdns > mdns.mcast.net.mdns: 0 PTR (QU)? _microsoft_mcc._tcp.local. (43)
0x0000: 4500 0047 8ee8 0000 0111 9118 c0a8 f801
0x0010: e000 00fb 14e9 14e9 0033 9292 0000 0000
0x0020: 0001 0000 0000 0000 0e5f 6d69 6372 6f73
0x0030: 6f66 745f 6d63 6304 5f74 6370 056c 6f63
0x0040: 616c 0000 0c80 01
13:19:56.924053 IP6 fe80::a3a8:cb23:5977:3ffe.mdns > ff02::fb.mdns: 0 PTR (QU)? _microsoft_mcc._tcp.local. (43)
0x0000: 6001 b819 0033 1101 fe80 0000 0000 0000
0x0010: a3a8 cb23 5977 3ffe ff02 0000 0000 0000
0x0020: 0000 0000 0000 00fb 14e9 14e9 0033 2578
0x0030: 0000 0000 0001 0000 0000 0000 0e5f 6d69
0x0040: 6372 6f73 6f66 745f 6d63 6304 5f74 6370
0x0050: 056c 6f63 616c 0000 0c80 01
13:19:56.986017 IP 192.168.248.131.55932 > 192.168.248.2.domain: 45900+ PTR? 251.0.0.224.in-addr.arpa. (42)
0x0000: 4500 0046 1b18 4000 4011 adb7 c0a8 f883
0x0010: c0a8 f802 da7c 0035 0032 721b b34c 0100
0x0020: 0001 0000 0000 0000 0332 3531 0130 0130
0x0030: 0332 3234 0769 6e2d 6164 6472 0461 7270
0x0040: 6100 000c 0001
13:19:56.993960 IP 192.168.248.131.60919 > 192.168.248.2.domain: 18650+ PTR? 251.0.0.224.in-addr.arpa. (42)
0x0000: 4500 0046 7f62 4000 4011 496d c0a8 f883
0x0010: c0a8 f802 edf7 0035 0032 721b 48da 0100

```

25. tcpdump -c 3: Captures only 3 packets and then automatically stops.

```

(root@kali)-[/home/kali]
# tcpdump -c 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:22:40.194514 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
13:22:40.238155 IP 192.168.248.131.46716 > 192.168.248.2.domain: 5919+ PTR? 2.248.168.192.in-addr.arpa. (44)
13:22:40.285407 IP 192.168.248.2.domain > 192.168.248.131.46716: 5919 NXDomain 0/1/0 (103)
3 packets captured
7 packets received by filter
0 packets dropped by kernel
(root@kali)-[/home/kali]
#

```

26. tcpdump -w my.pcap: Captures network packets and saves them to a file named my.pcap without displaying them on the screen.

```

(root@kali)-[/home/kali]
# tcpdump -w my.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
100 packets captured
100 packets received by filter
0 packets dropped by kernel
(root@kali)-[/home/kali]
# ls
10_201_38_28_nikto_fullscan.txt  flag.b64  kinju.pcap  Public
10_201_38_28_nikto_scan.txt    hashn.txt kinju.txt   result.csv
10_201_38_28_nmap              hash.txt  md5sum     staff.txt
198.168.248.1                 kinjal    Music      student.txt
198.168.248.2                 kinjal1   my.pcap    Templates
abc.txt                       kinjal12.pcapng  'New Graph (1).mtgl'  users.lst
Desktop                       kinjal20    'New Graph (2).mtgl'  Videos
Documents                     kinjal205.txt  nohup.out  'wireshark 2..pcap'
Downloads                     kinjal.txt   packetwireshartk.pcap 'wireshark 3..pcap'
ende.py                       kinju       Pictures   zphisher

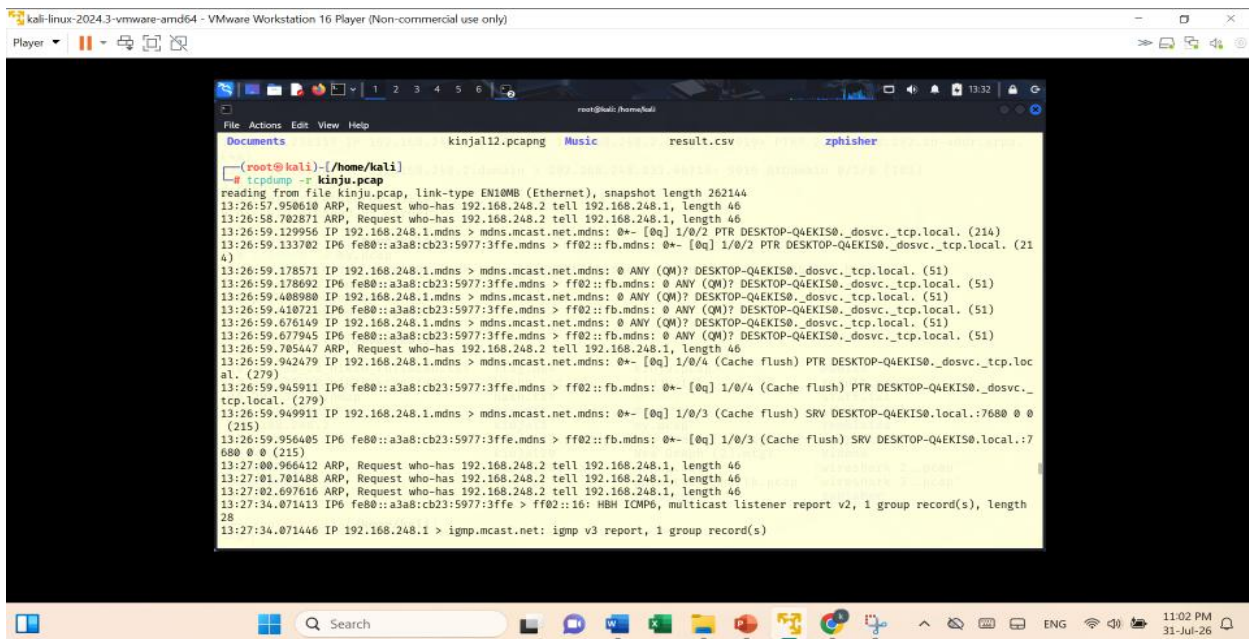
```

27. tcpdump -i eth0 -w kinju.pcap: Captures all network packets on the eth0 interface and saves them to a file named kinju.pcap without displaying them on the screen.

```
(root@kali)-[/home/kali]
# tcpdump -i eth0 -w kinju.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C164 packets captured
164 packets received by filter
0 packets dropped by kernel

(root@kali)-[/home/kali]
# ls
10_201_38_28_nikto_fullscan.txt  Downloads      kinjal20      my.pcap      staff.txt
10_201_38_28_nikto_scan.txt    ende.py       kinjal205.txt 'New Graph (1).mtgl' student.txt
10_201_38_28.nmap              flag.b64     kinjal.txt   'New Graph (2).mtgl' Templates
198.168.248.1                 hashn.txt   kinju        nohup.out    users.lst
198.168.248.2                 hash.txt    kinju.pcap   packetwireshartk.pcap Videos
abc.txt                       kinjal      kinju.txt    Pictures     'wireshark 2..pcap'
Desktop                      kinjal1     md5sum       Public       'wireshark 3..pcap'
Documents                   kinjal12.pcapng Music        result.csv   zphisher
```

28. tcpdump -r kinju.pcap: Read the file.



29. tcpdump -Q in: Captures only incoming (ingress) packets received by the interface.

Also use (out).

```
(root@kali)-[/home/kali]
# tcpdump -Q in
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:34:27.194400 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
13:34:27.291137 IP 192.168.248.2.domain > 192.168.248.131.37778: 30854 NXDomain 0/1/0 (103)
13:34:27.386270 IP 192.168.248.2.domain > 192.168.248.131.55960: 37390 NXDomain 0/1/0 (103)
13:34:27.429582 IP 192.168.248.2.domain > 192.168.248.131.41224: 49477 NXDomain 0/1/0 (105)
13:34:28.691718 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
13:34:29.693124 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
13:34:30.695662 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
13:34:32.493820 ARP, Reply 192.168.248.2 is-at 00:50:56:f2:4a:64 (oui Unknown), length 46
13:34:36.234998 ARP, Request who-has 192.168.248.2 tell 192.168.248.1, length 46
```

30. tcpdump portrange 80-100: Captures packets where the source or destination port is between 80 and 100.

```
(root@kali)-[/home/kali]
# tcpdump portrange 80-100
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

31. tcpdump -i eth0 -n dst or src host 192.168.248.1 and port 80: Captures incoming or outgoing HTTP (port 80) traffic to or from the host 192.168.248.1 on the eth0 interface, displaying IP addresses instead of resolving hostnames.

```
(root@kali)-[/home/kali]
# tcpdump -i eth0 -n dst or src host 192.168.248.1 and port 80
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

Conclusion:

I successfully explored and practiced various tcpdump commands to understand how network packets can be captured, filtered, displayed, and saved. I learned to capture traffic based on interfaces, hosts, networks, source/destination IPs, ports, and protocols, as well as how to save captured packets into a .pcap file for later analysis in Wireshark. This practical improved my understanding of network traffic analysis and packet capture, which are essential skills in cybersecurity, network troubleshooting, and digital forensics.

❖ Tshark

🚦 Definition:

TShark is a terminal-based network protocol analyzer. It serves as the command-line interface version of Wireshark, allowing you to capture live network traffic, read saved capture files, and automate packet analysis using scripts.

🚦 Installation:

Sudo apt install tshark

🚦 Practical:

1. tshark -v: Show Version.

```
(root@kali)-[/home/kali]
# tshark -v
Running as user "root" and group "root". This could be dangerous.
TShark (Wireshark) 4.2.5 (git v4.2.5 packaged as 4.2.5-1).

Copyright 1998-2024 Gerald Combs <gerald@wireshark.org> and contributors.
Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compiled (64-bit) using GCC 13.2.0, with GLib 2.80.2, with libpcap, with POSIX
capabilities (Linux), with libnl 3, with zlib 1.3.1, with PCRE2, with Lua 5.2.4,
with GnuTLS 3.8.5 and PKCS #11 support, with Gcrypt 1.10.3, with Kerberos (MIT),
with MaxMind, with nghttp2 1.61.0, with nghttp3 0.8.0, with brotli, with LZ4,
with Zstandard, with Snappy, with libxml2 2.9.14, with libsmi 0.4.8, with binary
plugins.

Running on Linux 6.8.11-amd64, with Intel(R) Core(TM) i3-1005G1 CPU @ 1.20GHz
(with SSE4.2), with 3645 MB of physical memory, with GLib 2.84.4, with libpcap
1.10.4 (with TPACKET_V3), with zlib 1.3.1, with PCRE2 10.46 2025-08-27, with
c-ares 1.31.0, with GnuTLS 3.8.9, with Gcrypt 1.11.0, with nghttp2 1.64.0, with
nghttp3 0.8.0, with brotli 1.1.0, with LZ4 1.10.0, with Zstandard 1.5.6, with
libsmi 0.4.8, with LC_TYPE=en_US.UTF-8, binary plugins supported.
```

2. tshark -h: Help, Show all Commands..

```
kali-linux-2024.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
File Actions Edit View Help
libsmi 0.4.8, with LC_TYPE=en_US.UTF-8, binary plugins supported.
# tshark -h
Running as user "root" and group "root". This could be dangerous.
TShark (Wireshark) 4.2.5 (git v4.2.5 packaged as 4.2.5-1)
Dump and analyze network traffic.
See https://www.wireshark.org for more information.
Usage: tshark [options] ...

Capture interface:
-i <interface>, --interface <interface>
    name or idx of interface (def: first non-loopback)
-f <capture filter>
    packet filter in libpcap filter syntax
-s <snaplen>, --snapshot-length <snaplen>
    packet snapshot length (def: appropriate maximum)
-p, --no-promiscuous-mode
    don't capture in promiscuous mode
-I, --monitor-mode
    capture in monitor mode, if available
-B <buffer size>, --buffer-size <buffer size>
    size of kernel buffer (def: 2MB)
-y <link type>, --linktype <link type>
    link layer type (def: first appropriate)
--time-stamp-type <type>
    timestamp method for interface
-D, --list-interfaces
    print list of interfaces and exit
-L, --list-data-link-types
    print list of link-layer types of iface and exit
--list-time-stamp-types
    print list of timestamp types for iface and exit
--update-interval
    interval between updates with new packets (def: 100ms)
```

3. tshark -D: List all available Network interfaces.

```
(root@kali)-[/home/kali]
# tshark -D
Running as user "root" and group "root". This could be dangerous.
1. eth0
2. any
3. lo (Loopback)
4. bluetooth0
5. bluetooth-monitor
6. nflog
7. nfqueue
8. dbus-system
9. dbus-session
10. ciscodump (Cisco remote capture)
11. dpauwmon (DisplayPort AUX channel monitor capture)
12. randpkt (Random packet generator)
13. sdjournal (systemd Journal Export)
14. sshdump (SSH remote capture)
15. udpdump (UDP Listener remote capture)
16. wifidump (Wi-Fi remote capture)

(root@kali)-[/home/kali]
#
```

4. tshark -i eth0: Captures live packets from the eth0 interface.

```
(root@kali)-[/home/kali]
# tshark -i eth0
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
1 0.000000000 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
2 1.452051410 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
3 2.000521121 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
4 3.000620985 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
5 8.998031141 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
6 9.996050670 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
7 10.997595650 192.168.248.1 → 192.168.248.255 BROWSER 243 Host Announcement DESKTOP-Q4EKIS0, Workstation, Server, NT Workstation
8 10.997596702 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
9 12.011150691 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
10 12.991013220 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
11 13.998118425 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
12 18.051032087 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
13 18.988102789 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
14 19.986031526 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
15 21.098956646 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
16 21.993606813 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
17 22.995420560 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
```

5. tshark -i eth0 -c 10: Captures only 10 packets and stops automatically.

```
(root@kali)-[/home/kali]
# tshark -i eth0 -c 10
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
1 0.000000000 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
2 4.052913662 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
3 4.989984364 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
4 5.987913101 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
5 7.100838221 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
6 7.995488388 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
7 8.997302135 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
8 36.136174292 fe80::f4d2:e529:b9db:4264 → ff02::2 ICMPv6 62 Router Solicitation
9 43.927350622 192.168.248.1 → 224.0.0.251 MDNS 256 Standard query response 0x0000 PTR DESKTOP-Q4EKIS0._dosvc._tcp.local SRV 0 0 7680 DESKTOP-Q4EKIS0.local TXT
10 43.935758849 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 276 Standard query response 0x0000 PTR DESKTOP-Q4EKIS0._dosvc._tcp.local SRV 0 0 7680 DESKTOP-Q4EKIS0.local TXT
10 packets captured

(root@kali)-[/home/kali]
#
```

6. tshark -r kinju.pcap: read pcap file.

```
(root@kali)-[/home/kali]
# tshark -r kinju.pcap
Running as user "root" and group "root". This could be dangerous.
1 0.000000 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
2 0.752261 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
3 1.179346 192.168.248.1 → 224.0.0.251 MDNS 256 Standard query response 0x0000 PTR DESKTOP-Q4EKIS0._dosvc._tcp.local SRV 0 0 7680 DESKTOP-Q4EKIS0.local TXT
4 1.183092 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 276 Standard query response 0x0000 PTR DESKTOP-Q4EKIS0._dosvc._tcp.local SRV 0 0 7680 DESKTOP-Q4EKIS0.local TXT
5 1.227961 192.168.248.1 → 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc._tcp.local, "QM" question
6 1.228082 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 113 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc._tcp.local, "QM" question
7 1.458370 192.168.248.1 → 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc._tcp.local, "QM" question
8 1.460111 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 113 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc._tcp.local, "QM" question
9 1.725539 192.168.248.1 → 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc._tcp.local, "QM" question
10 1.727335 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 113 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc._tcp.local, "QM" question
11 1.754837 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
12 1.991869 192.168.248.1 → 224.0.0.251 MDNS 321 Standard query response 0x0000 PTR, cache flush DESKTOP-Q4EKIS0._dosvc._tcp.local SRV, cache flush 0 0 7680 DESKTOP-Q4EKIS0.local TXT, cache flush A, cache flush 192.168.248.1 AAAA, cache flush fe80::a3a8:cb23:5977:3ffe
13 1.995301 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 341 Standard query response 0x0000 PTR, cache flush DESKTOP-Q4EKIS0._dosvc._tcp.local SRV, cache flush 0 0 7680 DESKTOP-Q4EKIS0.local TXT, cache flush A, cache flush 192.168.248.1 AAAA, cache flush fe80::a3a8:cb23:5977:3ffe
14 1.999301 192.168.248.1 → 224.0.0.251 MDNS 257 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-Q4EKIS0.local TXT, cache flush A, cache flush 192.168.248.1 AAAA, cache flush fe80::a3a8:cb23:5977:3ffe
15 2.005795 fe80::a3a8:cb23:5977:3ffe → ff02::fb MDNS 277 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-Q4EKIS0.local TXT, cache
```

7. tshark -f "port 80": Captures only packets matching the capture filter (port 80).

```
(root@kali)-[/home/kali]
# tshark -f "port 80"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

8. tshark -Y http: Display only HTTP packets using a display filter.

```
(root@kali)-[/home/kali]
# tshark -Y http
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

9. tshark -Y dns: Display only DNS packets.

```
(root@kali)-[/home/kali]
# tshark -Y dns
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

10. tshark -Y ip.addr==8.8.8.8: Display the packets where the ip is either source or destination.

```
(root@kali)-[/home/kali]
# tshark -Y ip.addr==8.8.8.8
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

11. tshark -y "tcp.port==443": Displays packets using TCP port 443 (HTTPS).

```
(root@kali)-[/home/kali]
# tshark -Y tcp.port==443
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

12. `tshark -T fields -e ip.src -e ip.dst`: Extracts and displays the source and destination IP addresses.

```
(root@kali)-[/home/kali]
# tshark -T fields -e ip.src -e ip.dst
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
192.168.248.1 → 224.0.0.251 MDNS 276 Standard query response 0x0000 PTR DESKTOP-Q4EKIS0._dosvc.
192.168.248.1 → 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc
192.168.248.1 → 224.0.0.251 MDNS 113 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc
192.168.248.1 → 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc
192.168.248.1 → 224.0.0.251 MDNS 113 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc
192.168.248.1 → 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-Q4EKIS0._dosvc
```

13. `tshark -t json`: Display packets data in json format.

```
(root@kali)-[/home/kali]
# tshark -t json

Running as user "root" and group "root". This could be dangerous.
tshark: Invalid time stamp type "json"; it must be one of:
"a"      for absolute
"ad"     for absolute with YYYY-MM-DD date
"adoy"   for absolute with YYYY/DOY date
"d"      for delta
"dd"     for delta displayed
"e"      for epoch
"r"      for relative
"u"      for absolute UTC
"ud"     for absolute UTC with YYYY-MM-DD date
"udoy"   for absolute UTC with YYYY/DOY date
```

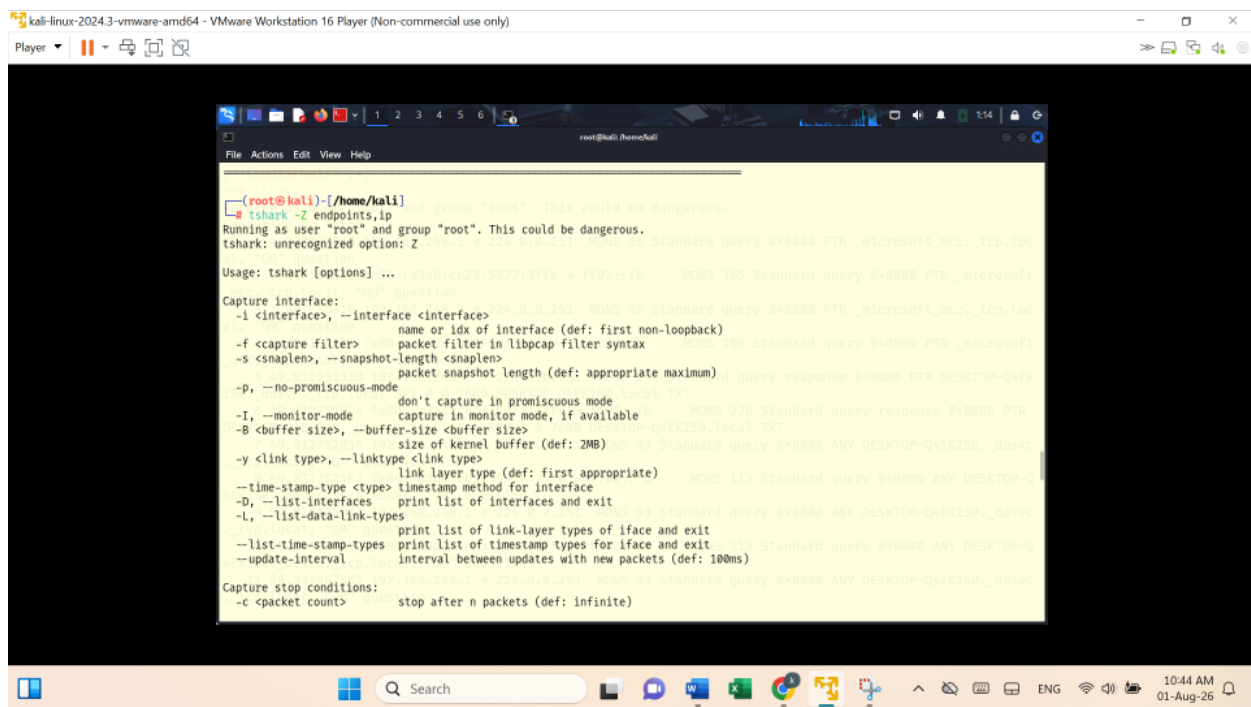
14. `tshark -z conv,tcp`: Display TCP conversation statistics.

```

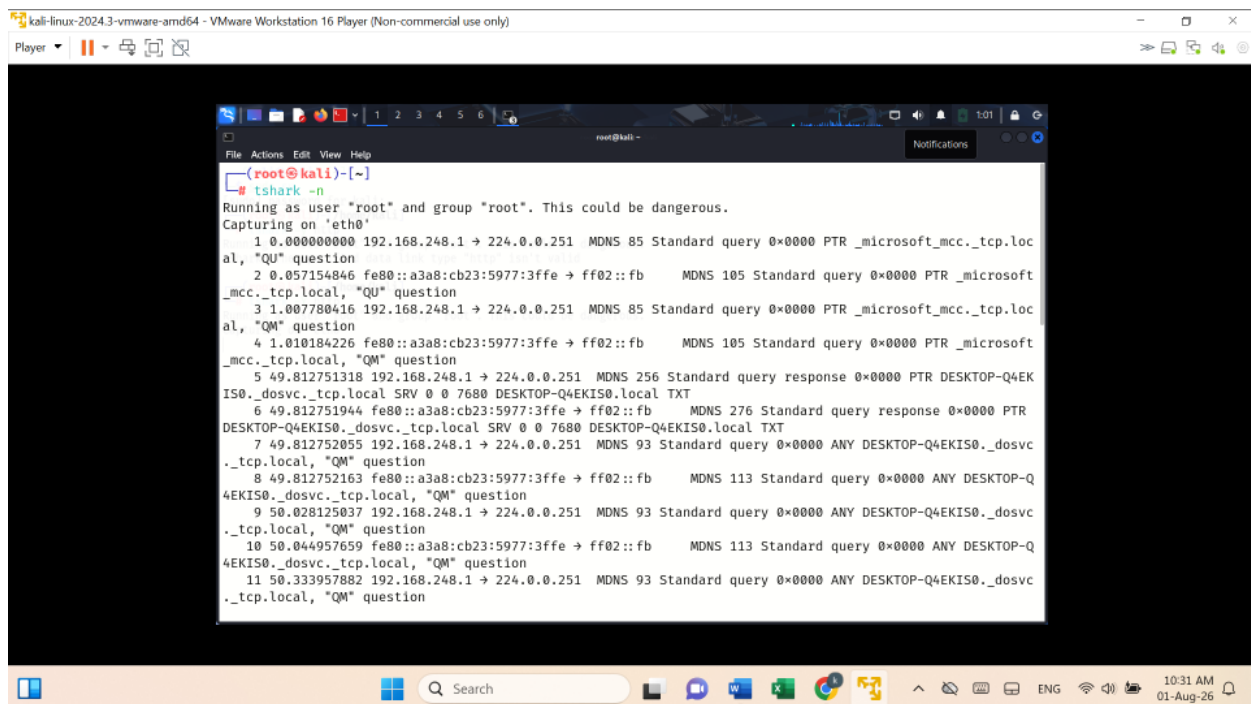
DESKTOP-Q4EKIS0._dosvr tcp,local SRV 0 0 7680 DESKTOP-Q4EKIS0.local TXT
1 (root@kali)-[/home/kali]
2 # tshark -z conv,tcp
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
3 1 0.000000000 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
4 2 0.027013508 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
5 3 2.011429706 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
6 4 3.010695896 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
7 5 8.584927231 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
8 6 9.507169341 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1
9 7 10.508242299 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.248.2? Tell 192.168.248.1

```

15. tshark -Z endpoints,ip: Displays ip endpoint statistics.



16. tshark -n: Disable hostname resolution and show only ip addresses.



Conclusion:

I successfully performed various TShark commands and learned how to capture and analyze network packets. I practiced using filters, reading PCAP files, and extracting network information. This practical improved my understanding of network traffic analysis and troubleshooting. It also strengthened my basic cybersecurity and packet analysis skills.